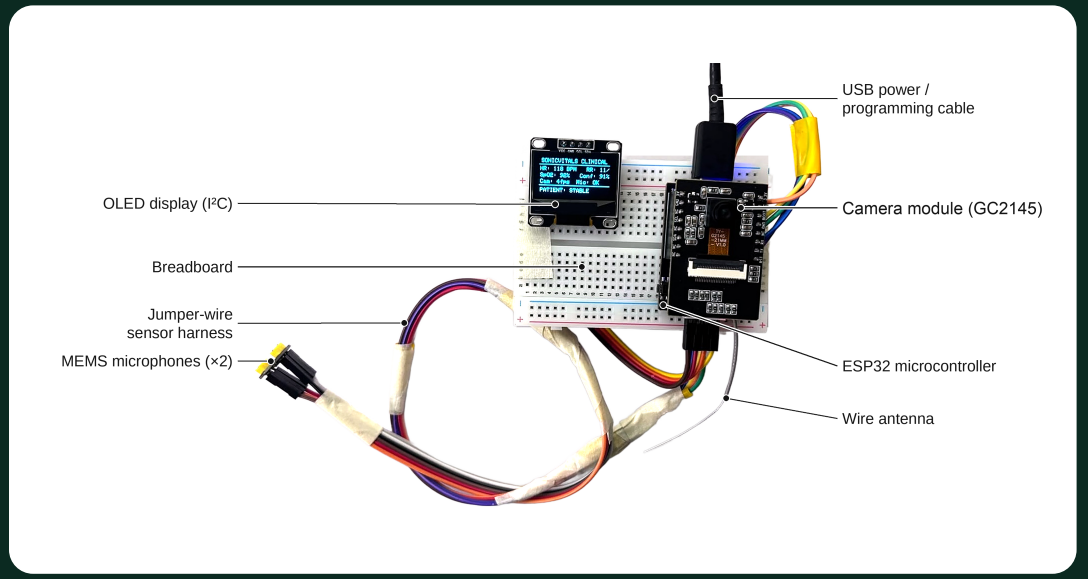
SonicVitals

Contactless heart-rate and breathing monitoring with a low-cost camera + microphone node

ESP32-S3 · rPPG · Heart sounds · Sensor fusion

SEM 5 Project · VIT · October 2026



Some people can't wear a sensor

Elderly and bedridden patients

People with dementia pull clips and straps off; nobody re-attaches them at night.

Fragile skin

Newborns and burn patients: adhesive sensors can injure the skin.

Shared screening

Health rooms and kiosks: quick checks for many people without a shared finger clip.

What already exists

Option	Cost	Limitation
Finger pulse oximeter	₹500–1,500	Needs contact
Smartwatch / band	₹2,000+	Must be worn and charged
Face-scan phone apps	App / licence	Closed; someone holds the phone
Radar / under-mattress	₹10,000+	Single purpose, closed

The gap: a cheap, open, fixed node that measures without contact and says when its reading can't be trusted.

What we built

Low-cost sensor node

ESP32-S3 + GC2145 camera + INMP441 microphone + OLED, about ₹1,500 of parts, streaming over Wi-Fi.

Two sensors, one clock

Every frame and audio block carries a microsecond timestamp from the same ESP32 clock, so camera and sound line up.

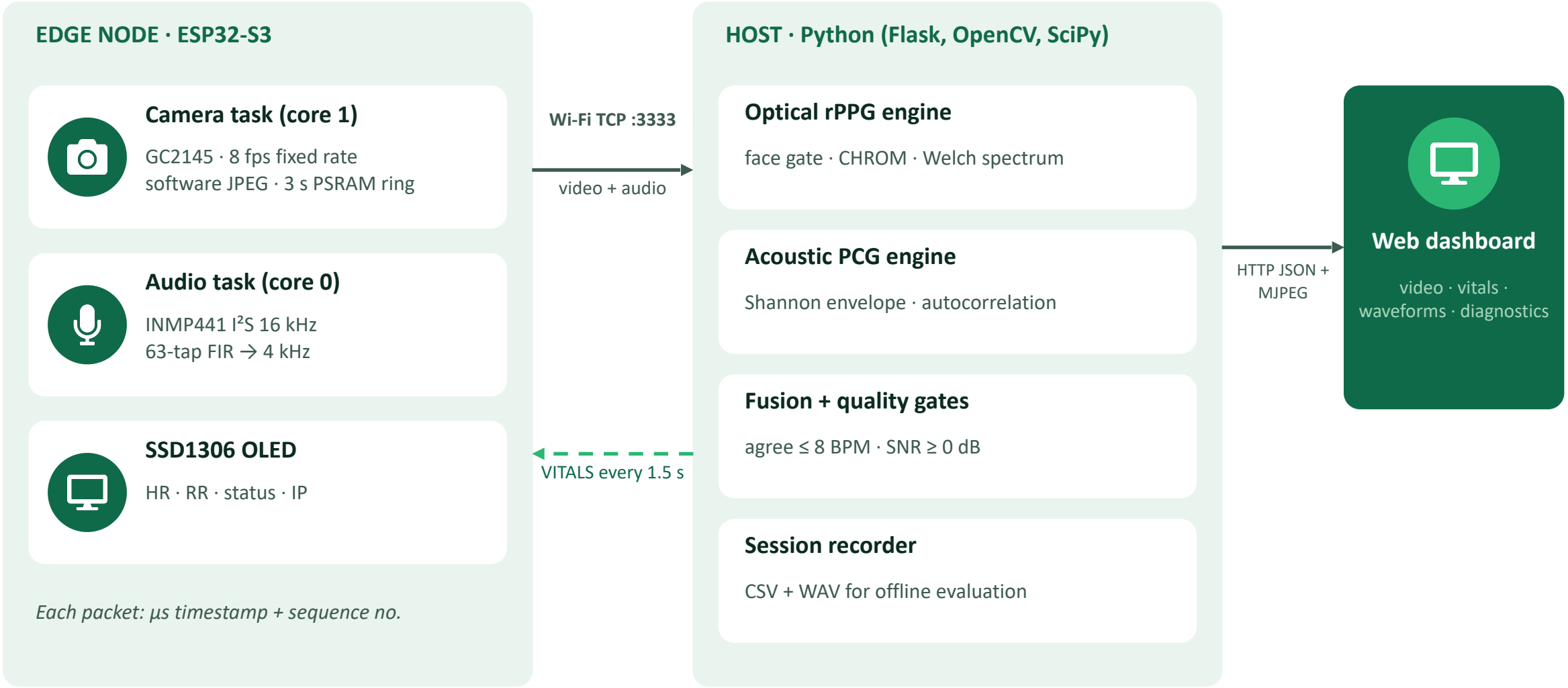
Fusion with an agreement check

Camera HR and heart-sound HR are averaged by quality only if they agree within 8 BPM.

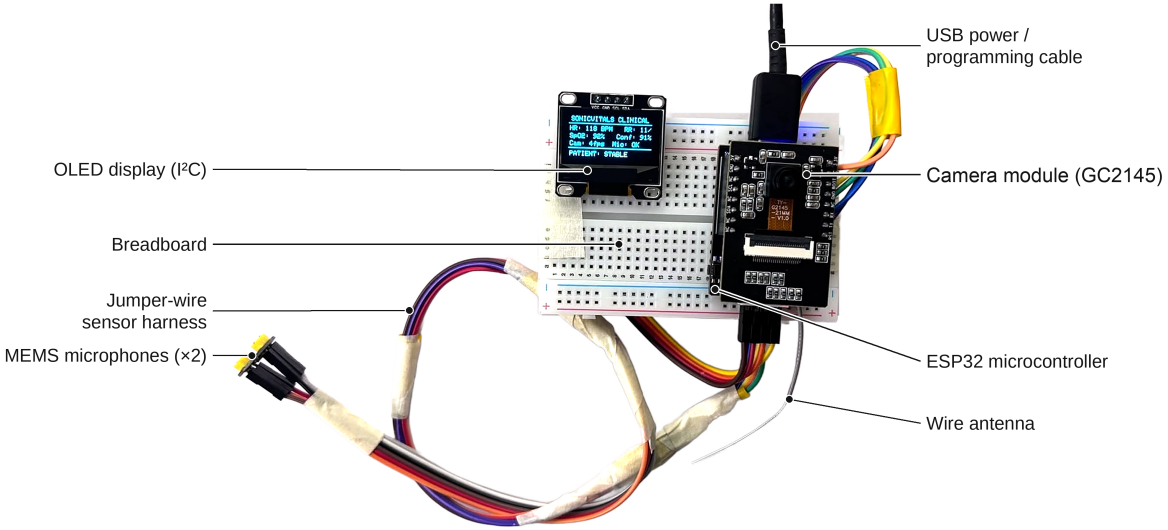
Honest quality gating

A reading is shown only if its signal clearly beats noise; otherwise the display shows "--".

System architecture



Engineering the node: problems we hit



ESP32-S3: 2 cores at 240 MHz, 8 MB PSRAM, Wi-Fi
GC2145: 2 MP camera, used at 240×160
INMP441: 24-bit I²S MEMS microphone
SSD1306: 128×64 OLED over I²C

Mic pins = USB-serial pins

GPIO43/44 carry both → stream over Wi-Fi instead

No hardware JPEG on GC2145

Capture RGB565, compress in software, fixed 8 fps

Auto-exposure hides the pulse

Lock exposure, gain, white balance after 3 s

Hotspot gives only ~70 kB/s

Audio 16 → 4 kHz (FIR), 3 s PSRAM frame buffer

From pixels and sound to heart rate

OPTICAL · camera

Face gate
Haar cascade + 5 checks

→

Skin colour
forehead + cheeks mean R,G,B

→

Resample
timestamps → 15 Hz grid

→

CHROM
$S = X - \alpha Y$ cancels motion

→

Band-pass
0.75–2.5 Hz (45–150 BPM)

→

HR
Welch peak, tracked
shown if SNR ≥ 0 dB

ACOUSTIC · chest microphone

Band-pass
25–400 Hz heart sounds

→

Resample
1 kHz

→

Envelope
Shannon energy $-x^2 \log x^2$

→

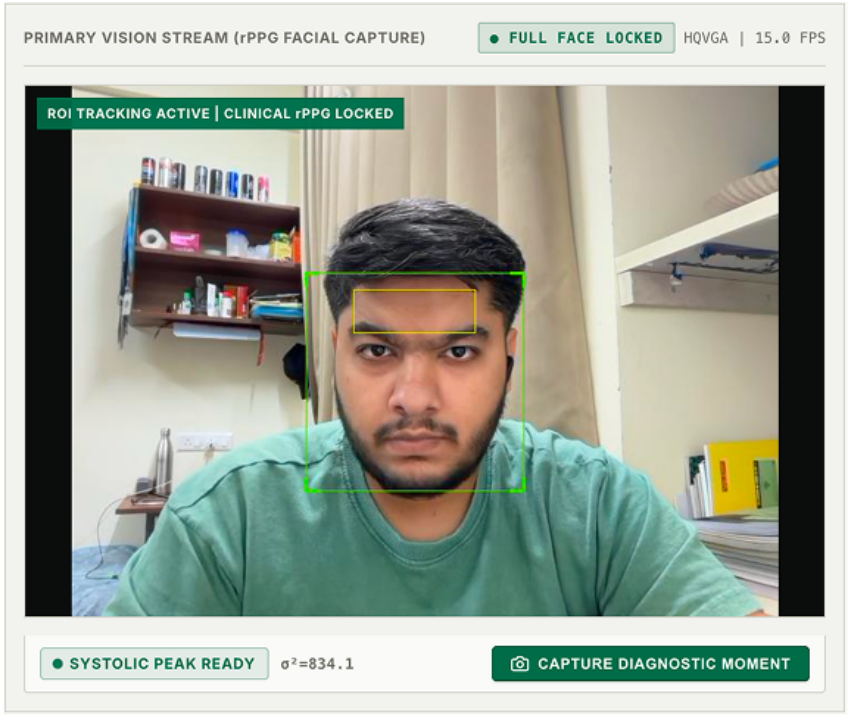
HR
autocorrelation, $60/\tau$
shown if $\rho \geq 0.3$

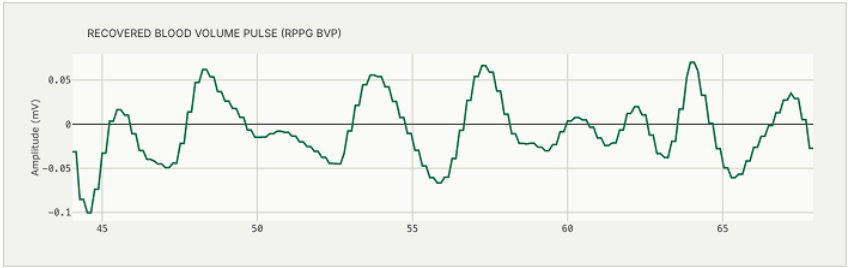
Breathing
camera green 0.15–0.40 Hz → RR

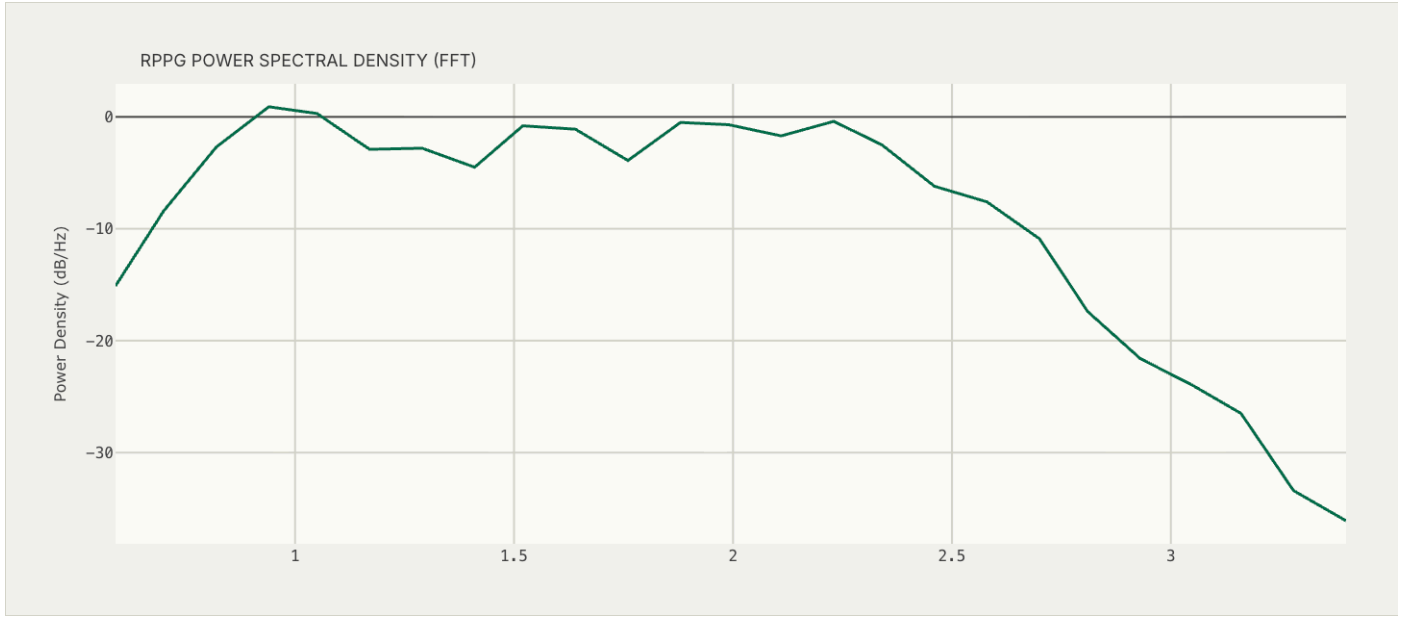
Fusion

If camera and sound agree within 8 BPM: HR = (w_cam·HR_cam + w_snd·HR_snd) / (w_cam + w_snd), weighted by signal quality. If they disagree, use the better channel and flag a conflict. Brief disturbances (≤ 2 s) are bridged; longer ones hold the last value for 15 s.

The live dashboard







1 Face gate + skin ROI

Green box = face passed all 5 checks; yellow = skin patch used for the pulse.

2 Recovered pulse wave

Band-passed CHROM signal; each hump is one heartbeat.

3 Pulse spectrum

Highest peak in 0.75–2.5 Hz gives HR = 60 × f_o.

What we measured

Frame-timing jitter (SD) — lower is better

Wi-Fi, first firmware



310 ms

USB-serial 1 Mbaud (mic off)



15.5 ms

Wi-Fi, fixed-rate capture

0.4 ms

775× steadier after moving capture into a fixed-rate camera task with timestamps taken at capture.

8.0

frames / s received

evenly spaced, Wi-Fi

0

frames or audio blocks lost

30 s run, checked by sequence no.

±2 BPM

HR on synthetic signals

at 52, 78 and 128 BPM

≈5 %

noise passing the gate

pure-noise test, so "--" not fake HR

Not yet measured: HR accuracy against an ECG chest strap (planned study).

Limitations and next steps

LIMITATION	NEXT STEP
No reference validation yet	Compare with an ECG chest strap: Bland–Altman, AAMI ±5 BPM
Needs light	Add an IR LED + IR-sensitive camera for night monitoring
Motion and lighting changes	Landmark face tracking; colour means computed on the node before JPEG
Heart sounds weak through air	Contact or stethoscope-coupled microphone on the chest
Colour indices uncalibrated	Colour reference card and fixed lighting; exploratory only

Takeaways



Contactless HR and breathing from a ≈₹1,500 open node, with no phone or wearable.



Camera and heart-sound data on one clock, fused only when the two agree.



Quality gates show "--" instead of a confident wrong number.

Thank you · Questions?